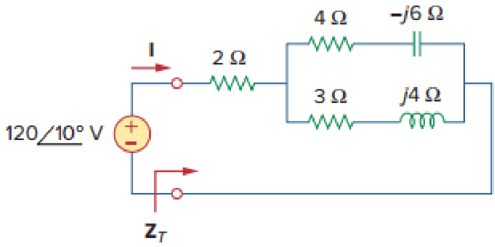
Chapter 9, Problem 65P

Problem

Determine **Z***T* and **I** for the circuit in Fig. 9.72.

Figure 9.72:



Step-by-step solution

Step 1 of 1

$$Z_T = 2 + (4 - j6) \parallel (3 + j4)$$

$$Z_T = 2 + \frac{(4 - j6)(3 + j4)}{7 - j2}$$

$$Z_T = 6.83 + j1.094\Omega$$

$$= 6.917\angle 9.1\,\Omega$$

$$I = \frac{V}{Z_T} = \frac{120\angle 10^\circ}{6.917\angle 9.1\,\Omega}$$

$$= 17.35\angle 0.9^\circ\ A\ .$$